每日 MIR 論文精選 - 2026-05-24

優先序：MeanAudio / text-to-audio generation → karaoke-jp → genre/tagging。

Generated at 2026-05-24 01:36:59 Asia/Taipei.

#1 MeanAudio / text-to-audio 主線 新且重要

Live Music Diffusion Models: Efficient Fine-Tuning and Post-Training of Interactive Diffusion Music Generators

Interactive streaming music generation promises the use of generative models for live performance and co-creation tha...

text-to-audio/music

2026-05-21 新穎性 3 天 mean 4 / kara 0

摘要

Interactive streaming music generation promises the use of generative models for live performance and co-creation that is impossible with offline models. However, SOTA models exist in the discrete-AR regime, requiring industrial levels of compute for both training and inference. In this work, we investigate whether audio diffusion models, with their wide support in the open-source community but non-streaming bidirectional nature, can be repurposed efficiently into interactive models accessible on consumer hardware. By taking a critical look at the modern pipeline for block-wise outpainting diffusion, we identify critical inefficiencies during inference that result in strictly worse computational efficiency than their discrete-AR counterparts. We propose Live Music Diffusion Models (LMDMs), a simple modification of the generative diffusion process that recovers, and then outperforms, the inference complexity of the discrete Live Music Models (LMMs) through block-wise KV Caching. Unlike LMMs, LMDMs further enable stable post-training alignment through our

可行性

可行性高：若方法牽涉資料、caption、評估或 sampling，可以拆成一個小 ablation 先驗證。

novel ARC-Forcing paradigm, reducing error accumulation without any explicit RL or reward models. We demonstrate the application of LMDMs in a number of creative domains, including text-conditioned generation, sketch-based music synthesis, and jamming. We finally show how LMDMs can be used as a generative instrument in a real artist-AI collaboration, utilizing LMDMs as a "generative delay" to transform musicians' improvisation live for variable timbral effects while running locally on a consumer gaming laptop.

主要貢獻（快速驗證）

先看問題定義、baseline、評估資料、音訊樣本或程式碼是否可信，再決定是否深讀。

和你的研究/專案的關係

優先檢查它能否影響 MeanAudio / text-to-audio 生成的資料與 caption 策略、訓練穩定性、評估方法、速度/RTF、codec 或 reward alignment。

下一步

如果有新方法，優先抽成單一變因小實驗；如果是評估論文，先檢查它能否補上 CLAP/text-audio similarity 的盲點。

注意事項

若只依賴單一自動化指標，尤其 text-audio similarity 類，請用實際樣本、ablation 與多指標交叉檢查。

作者 Zachary Novack, Stephen Brade, Haven Kim, Hugo Flores García, Nithya Shikarpur, Chinmay Talegaonkar, Suwan Kim, Valerie K. Chen, Julian McAuley, Taylor Berg-Kirkpatrick, Cheng-Zhi Anna Huang

來源 https://arxiv.org/abs/2605.22717v1

#2 MeanAudio / text-to-audio 主線 新且重要

Academic Text-to-Music Grand Challenge: Datasets, Baselines, and Evaluation Methods

This paper presents an overview and the technical framework of the ICME 2026 Grand Challenge on Academic Text-to-Musi...

text-to-audio/music caption/data evaluation

2026-05-20 新穎性 4 天 mean 4 / kara 0

摘要

This paper presents an overview and the technical framework of the ICME 2026 Grand Challenge on Academic Text-to-Music Generation (ATTM). Despite the rapid progress in text-to-music generation (TTM) systems, the field is currently dominated by models trained on massive proprietary datasets with industrial-scale computational resources, creating a significant barrier for academic research. To address this, the ATTM Challenge establishes a fair-play benchmark that requires participants to train generative models strictly from scratch using a standardized, CC-licensed subset of the MTG-Jamendo dataset containing only instrumental music. The challenge is divided into two tracks: the Efficiency Track (limited to 500M parameters) and the Performance Track (no parameter limit). Submissions are evaluated through a multi-stage process involving objective metrics, including Frechet Audio Distance, CLAP score, and a novel Concept Coverage Score (CCS), followed by a subjective listening test. By providing open-source baselines, preprocessing pipelines, reference captions,

可行性

可行性高：若方法牽涉資料、caption、評估或 sampling，可以拆成一個小 ablation 先驗證。

and public evaluation code for computing FAD and CLAP, this challenge aims to facilitate and promote TTM research in academic contexts.

主要貢獻（快速驗證）

先看問題定義、baseline、評估資料、音訊樣本或程式碼是否可信，再決定是否深讀。

和你的研究/專案的關係

優先檢查它能否影響 MeanAudio / text-to-audio 生成的資料與 caption 策略、訓練穩定性、評估方法、速度/RTF、codec 或 reward alignment。

下一步

如果有新方法，優先抽成單一變因小實驗；如果是評估論文，先檢查它能否補上 CLAP/text-audio similarity 的盲點。

注意事項

若只依賴單一自動化指標，尤其 text-audio similarity 類，請用實際樣本、ablation 與多指標交叉檢查。

作者	Fang-Chih Hsieh, Wei-Jaw Lee, Chun-Ping Wang, Hung-yi Lee, Hao-Wen Dong, Yi-Hsuan Yang
來源	https://arxiv.org/abs/2605.21538v1

今天的 take-away

先用「能不能轉成你下一個小實驗」來讀：資料/caption、評估、速度/RTF、可控性與對齊。如果論文依賴相似度類指標，請補實際聆聽、多指標與 ablation，避免被漂亮分數帶偏。

Generated by the local MIR paper automation.